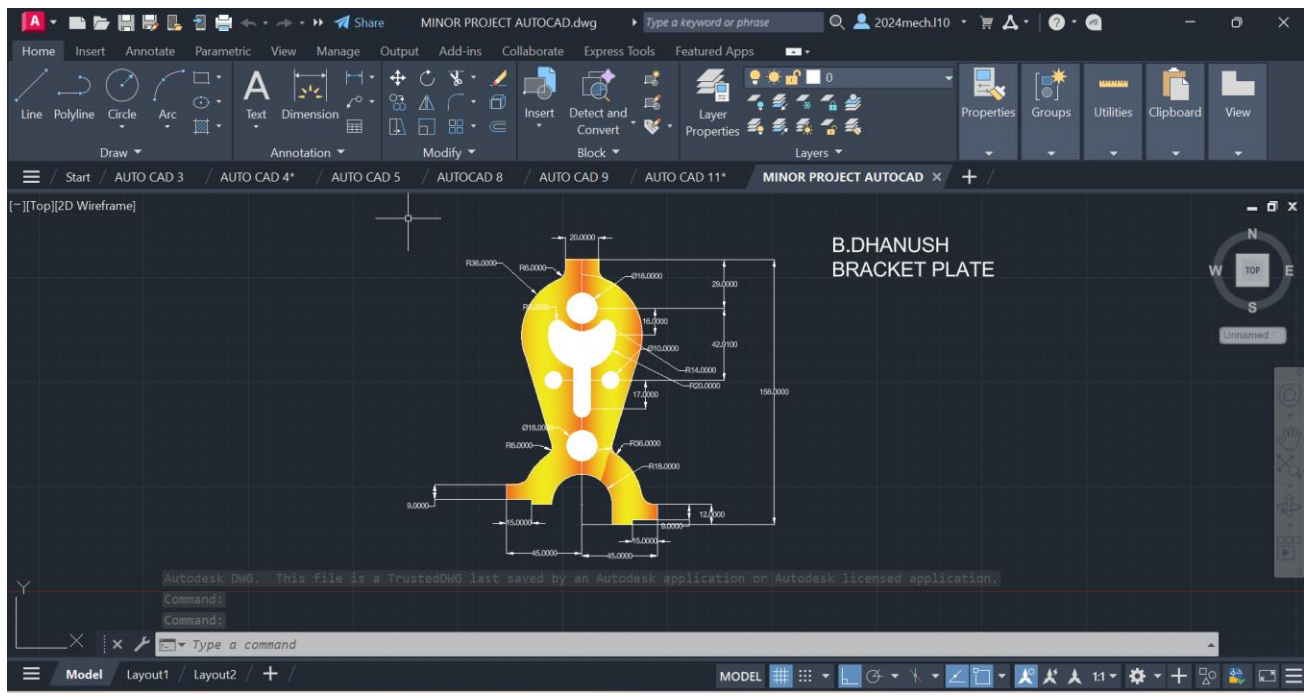


Projects

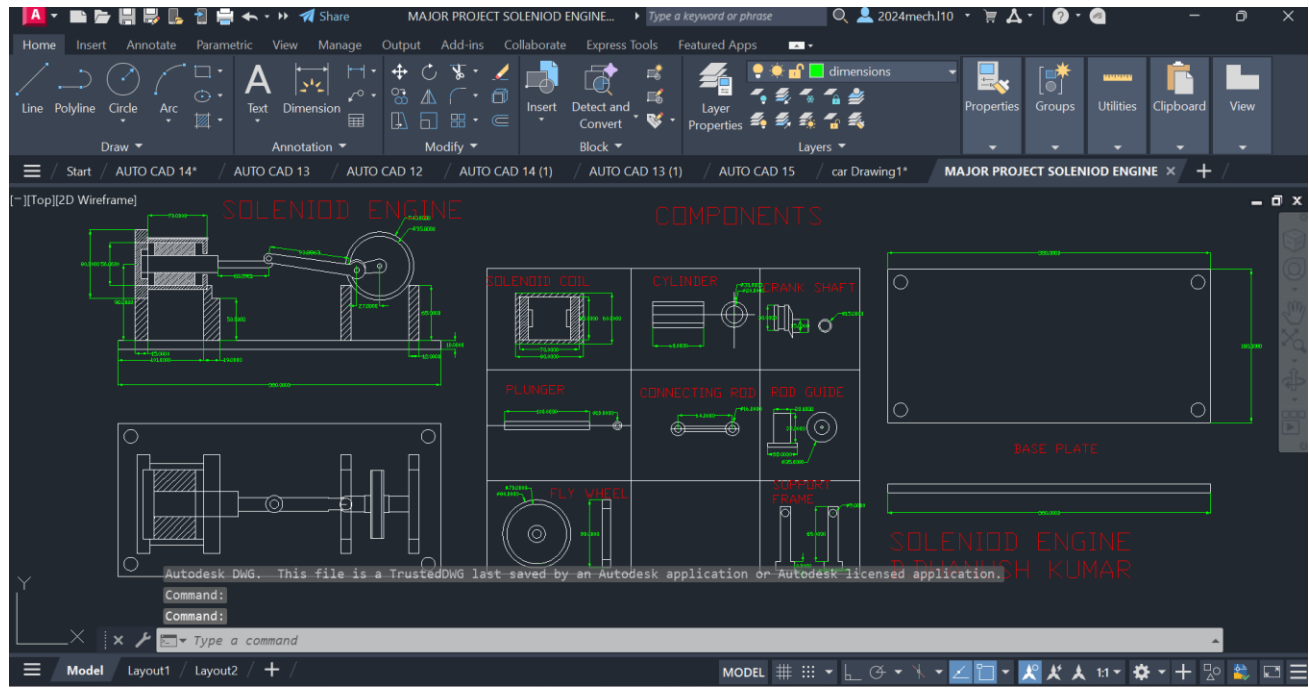
Project 1:



Procedure :

1. First of all open autocad software in the laptop or in desktop .
2. After opening, click on new the above page will appear in screen.
3. If you don't need graphical page click on 'F7' the graphical page will disappear in screen.
4. After that press on letter 'L' for line command or you can directly click on line command which will be available in command box.
5. Using line command draw the diagram with required dimensions.
6. use the circle,trim and arc commands, where it is needed.
7. change the dimensions and units by clicking letters 'Dim' for 'Dimensions' and 'U' for 'Units' after clicking change the units.
8. After drawing the diagram save it in laptop or in desktop.

Project 2:

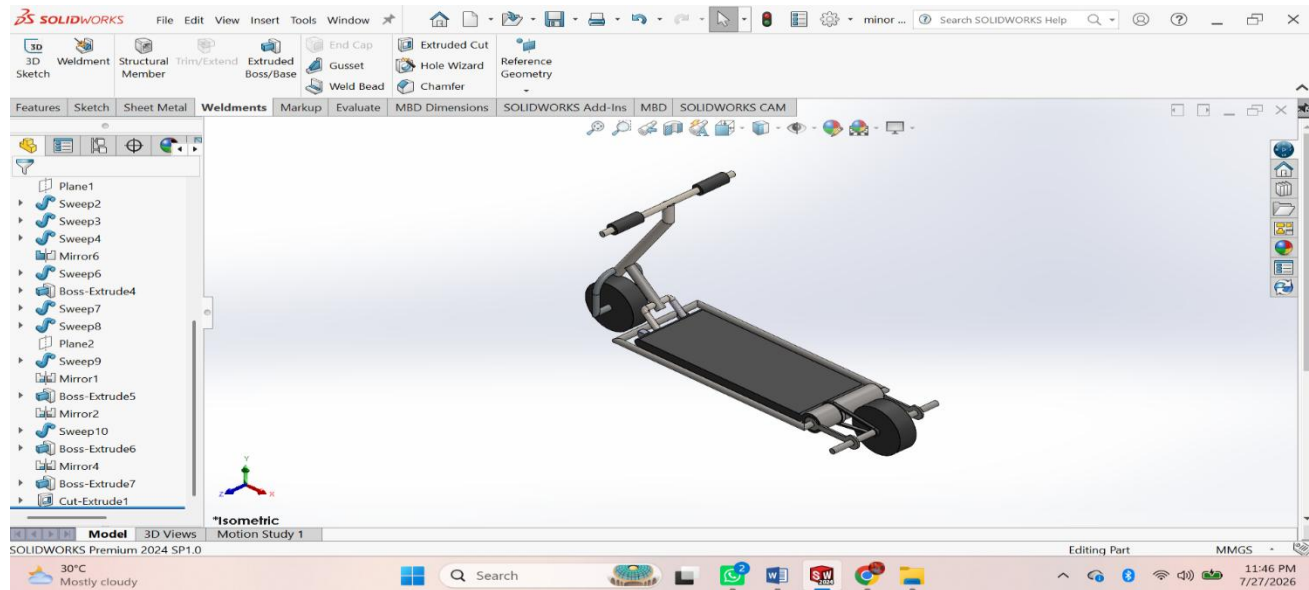


Procedure :

1. First of all open autocad software in the laptop or in desktop .
2. After opening, click on new the above page will appear in screen.
3. If you don't need graphical page click on 'F7' the graphical page will disappear in screen.
4. After that press on letter 'L' for line command or you can directly click on line command which will be available in command box.
5. Using line command draw the diagram with required dimensions.
6. use the circle, trim and arc commands, where it is needed.
7. change the dimensions and units by clicking letters 'Dim' for 'Dimensions' and 'U' for 'Units' after clicking change the units.
8. After drawing the diagram save it in laptop or in desktop.

SOLIDWORKS

PROJECT 1



Procedure

1. First of all open solidworks software in the laptop or in desktop.
2. After opening, click on new the above page will appear in screen.
3. After that click on front view the sketch chuck box will be appear click on it for further step.
4. In that page a small rectangular box will appear at above the date and time box, there change the units from 'MKS' to 'MMGS' units.
5. Using 'Circle' command draw the diagram with required dimensions by giving alignments to circle.
6. After drawing open 'Extrude' command and extrude upto required dimensions.
7. After that click on top of the extrude part, a sketch box will be appear click on it.
8. Again Use Line command and draw the next swept part.

9. After drawing, by using 'swept' command draw the thread upto required dimensions.
10. After drawing the diagram save it in laptop or in desktop.

PROJECT 2

Procedure

- **Create all Parts** – Base, Frame, Crank, Shaft, Connecting Rod, Wing Link, Wing, Pins, Bushes, and Fasteners.
- **Save** each part as **.SLDPRT**.

Assembly

- **File → New → Assembly.**
- **Insert Base** and **Fix** it.
- **Insert Frame** – Apply **Coincident Mate**.
- **Insert Shaft** – Apply **Concentric Mate**.
- **Insert Crank** – Apply **Concentric** and **Coincident Mates**.
- **Insert Connecting Rod** – Apply **Concentric Mate**.
- **Insert Wing Link** – Apply **Concentric Mate**.
- **Insert Left and Right Wings** – Apply **Hinge (Concentric) Mates**.
- **Insert Pins, Bushes, and Fasteners** – Apply **Concentric** and **Coincident Mates**.
- **Apply Mechanical Mates** (Hinge/Limit Angle/Gear Mate if required).
- **Check Interference – Evaluate → Interference Detection.**
- **Create Motion Study** – Add **Rotary Motor** to the crankshaft and run the simulation.
- **Save the Assembly** (.SLDASM) and create the **Assembly Drawing** with **BOM and Balloons**.